

Netflix Movie Performance Analysis

Business Requirement

Business Scenario

Netflix wants to understand which genres and movies perform best on its platform based on audience popularity and user ratings. By analyzing content performance, the company aims to identify trending genres, highly rated movies, and audience preferences to improve content recommendations and future content acquisition decisions.

As a **Junior Data Analyst**, your responsibility is to analyze the Netflix movie dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to identify the most popular genres, highest-rated movies, language preferences, and content trends.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and provide meaningful recommendations.

Project Objective

Analyze the Netflix movie dataset to:

- Understand the popularity of different movie genres.
 - Identify the highest-rated and most popular movies.
 - Analyze audience engagement using ratings and vote counts.
 - Study movie release trends over time.
 - Generate business insights and recommendations.
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Dataset Description

The dataset contains information about **Netflix movie titles**, including release date, popularity score, audience ratings, vote count, original language, genre, and movie overview.

Dataset Features

Column	Description
Release Date	Movie Release Date
Title	Movie Title
Overview	Short Description of the Movie
Popularity	Popularity Score
Vote Count	Total Audience Votes
Vote Average	Average Audience Rating
Original Language	Original Language of the Movie
Genre	Movie Genre
Poster URL	Movie Poster Link

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
 - Load the dataset using Pandas.
 - Display the first five records.
 - Display the last five records.
 - Find the number of rows and columns.
 - Display all column names.
 - Check the data types of each column.
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Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
 - Check duplicate records.
 - Display summary statistics.
 - Find the number of unique genres.
 - Find the number of unique original languages.
 - Find the total number of movies.
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Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
 - Handle missing values.
 - Convert **Release Date** into Date format.
 - Create a new column named **Release Year**.
 - Ensure **Popularity**, **Vote Count**, and **Vote Average** have the correct data types.
 - Remove unnecessary spaces from text columns.
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Business Questions

Answer the following questions using Python and Pandas.

Movie Analysis

- How many movies are available in the dataset?
 - Which movies have the highest popularity scores?
 - Which movies have the highest audience ratings?
 - Display the Top 10 most popular movies.
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Genre Analysis

- Which genres appear most frequently?
 - Which genres have the highest average popularity?
 - Which genres have the highest average audience ratings?
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Audience Analysis

- Which movies received the highest number of audience votes?
 - Which movies have the highest average ratings?
 - Is there any relationship between popularity and vote count?
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Language Analysis

- Which original language appears most frequently?

- Which language has the highest average popularity?
 - Which language has the highest average audience ratings?
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Release Year Analysis

- How many movies were released each year?
 - Which release year contains the highest number of movies?
 - Which release year has the highest average popularity?
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Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

- Top 10 Movie Genres (Bar Chart)
- Top 10 Most Popular Movies (Horizontal Bar Chart)
- Movies Released by Year (Line Chart)
- Popularity Distribution (Histogram)
- Original Language Distribution (Pie Chart)

Optional Chart

- Top 10 Movies by Vote Count (Bar Chart)
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Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which genres are most popular among viewers?
 - Which movies received the highest audience engagement?
 - Which languages dominate the Netflix movie collection?
 - Are newer movies generally more popular?
 - Do highly rated movies also receive more audience votes?
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Business Recommendations

Provide **at least 3 recommendations** for Netflix.

Example recommendations:

- Invest more in genres that consistently receive high popularity and ratings.
 - Recommend highly rated movies to improve user engagement.
 - Acquire more content in languages with growing audience demand.
 - Promote recently released, high-performing movies on the platform.
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Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
 - Cleaned Dataset (if modified)
 - Report
 - 3 Business Recommendations
 - 5-minute PowerPoint Presentation
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Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	100